
Validating simulation-based reporting rules: coverage, false positives and unstable thresholds

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Abstract

1 A 50-cell reporting rule supported by point-estimate recovery fails direct interval
2 evaluation in two colorectal single-cell cohorts. At its reporting boundary, a paired
3 simulation study finds 84.5–91.0% coverage of the fixed source-pool effect and
4 12.0–30.0% false-positive rates for nominal 95% intervals. Scoring the same
5 intervals against a sampled reference instead of the generating-pool parameter
6 changes the assessment: at 800 cells, inclusion falls from 99.0–100.0% to 90.5–
7 94.0% when the target is fixed. An exact variance calculation explains how the
8 sampled reference removes part of the error being evaluated. Controlled grid
9 comparisons also identify three of eight seed streams that fail at a larger count after
10 first qualifying. Greater relative expression variance helps explain weaker detection
11 in the second cohort. This self-audit documents how target choice, sparse-sample
12 interval behavior and threshold selection affected an intervention simulation. A
13 synthetic external-control example further shows that a reference for the wrong
14 population can favor an estimator whose median effect exceeds an illustrative
15 decision threshold while the intended target falls below it. The resulting evaluation
16 separates fixed-pool coverage, null rejection, effect detection and abstention, with
17 explicit source populations and Monte Carlo uncertainty.

18 1 Motivation: recovery and decision-rule validation

19 A reporting rule can withhold unreliable estimates only if its thresholds are calibrated to the uncertainty
20 of the reported quantity. We examine a 50-cell rule in colorectal single-cell analysis through three
21 empirical tests: coverage against a fixed empirical source-pool target, rejection under the null, and
22 stability of first-crossing candidates. The study is a self-audit of a previously overstated validation
23 claim in an unpublished project. Its contribution is the measured behavior of this rule and the
24 reproducible target comparison, rather than a new principle connecting point and interval accuracy.

25 Our motivating application separates two explanations for differentiation-marker loss in colorectal
26 cancer: fewer marker-producing cells (a *compositional* change), or lower output from the cells that
27 remain (an *intrinsic* change). These explanations suggest different biological interpretations. An
28 empirical comparison of per-cell means requires the relevant population to be observed in both tissues.
29 Let f_N, f_T denote the relevant cell fractions in reference and diseased tissue, and m_N, m_T their
30 mean per-cell output. A reference-based decomposition, related to demographic standardisation and
31 Oaxaca–Blinder decompositions [Kitagawa, 1955, Oaxaca, 1973, Blinder, 1973], is

$$\underbrace{(f_T - f_N)m_N}_{\text{compositional}} + \underbrace{f_N(m_T - m_N)}_{\text{intrinsic}} + \underbrace{(f_T - f_N)(m_T - m_N)}_{\text{interaction}} = f_T m_T - f_N m_N. \quad (1)$$

32 The interaction is reported separately. If no relevant cells are observed in an arm, its sample mean is
33 undefined; the identity cannot supply the missing mean. In a synthetic population with no such cells,

the corresponding conditional mean itself has no referent. Substituting zero in either case turns an unavailable contrast into an apparent effect.

The implemented reporting rule returns an estimate at $n \geq 50$, flags a wide interval at $20 \leq n < 50$, and returns NOT ESTIMABLE below 20, where n is the relevant-cell count in diseased tissue. Unavailable reported contrasts are stored as None, not zero. The diagnostic oracle calculations examined below also retain raw arithmetic before this reporting gate. We evaluate the boundary directly and retain pre-gate intervals at smaller counts to diagnose the curve’s behavior.

Relation to existing validation methods. Simulation-study methodology already separates estimands, methods and performance measures [Burton et al., 2006, Morris et al., 2019]. Generator–analysis alignment is discussed in congeniality [Meng, 1994], plasmode and semi-synthetic benchmarks [Franklin et al., 2014, Curth et al., 2021], interventional evaluation [Gentzel et al., 2019] and the inverse crime [Wirgin, 2004, Kaipio and Somersalo, 2007]; generators can also disadvantage good estimators [Shaw et al., 2026]. We apply these distinctions to the recorded reporting-rule decision. For clinical trials, external-control validation already examines bias and type-I error using held-out studies [Ventz et al., 2019]. Prognostic adjustment with historical data retains trial randomisation and evaluates precision gains [Schuler et al., 2022]; target-trial emulation states the causal question and design explicitly [Hernán and Robins, 2016]. These methods motivate the clinical-simulation validation questions in Section 3. The empirical study uses single-cell pools; a separate synthetic external-control experiment illustrates target-population mismatch. Neither validates a patient world model. Simulation-based calibration evaluates posterior ranks [Talts et al., 2018, Modrák et al., 2025]. Appendix E provides further context.

2 Calibration study and results

The claim being re-evaluated. An internal assessment (Artifact A) treated the 50-cell boundary as validated because recovery ratios were 0.85–1.08 above it and 1.18 in the 20–49-cell band, using six draws per setting at 1,200 cells per arm. The design record (Artifact B) instead required at least 90% interval coverage and 80% exclusion of zero at a halving of expression. Appendix E and the anonymized supporting record make this mismatch inspectable.

2.1 Simulation and inferential targets

The design document fixed a detectable halving of per-cell output ($s = 0.5$), coverage of the target in at least 90% of defined outputs, and discrimination of at least 80%, defined as the probability that the interval excludes zero at that effect. The implementation scored coverage against a drawn-reference benchmark; we retain that score and add coverage against the generating pool mean. The candidate *ok* is the first evaluated count or bin meeting both criteria; *wide* is the first meeting coverage alone. The committed values are 50 and 20. The first-crossing definition does not require larger counts to qualify.

The calibration uses paired subsets of the published colorectal atlas [Lee et al., 2020]: ten patients from GSE132465 (SMC) and six from GSE144735 (KUL3). We analyse GUCA2A in the differentiated population under fixed lineage labelling. Each draw samples, with replacement, 2,000 cells per arm from an eligible two-patient pool; the reference mature fraction is $f_N = 0.40$. The implemented pool combines cells from both tissues before drawing either synthetic arm. Because the design document did not specify tissue eligibility, we also evaluate a reference-tissue-only pool. Both arms sample the same eligible pool, with an imposed expression shift. A nominal 95% percentile interval resamples mature cells in both arms, with fractions held fixed and 200 inner draws unless stated otherwise. Undefined intervals are excluded from rate denominators and reported as abstentions (Appendix B).

The stored “parametric” intrinsic reference is

$$i_{\text{req}}(D_N; s) = f_N(s - 1)\bar{Y}_N, \quad (2)$$

where $\bar{Y}_N$ is the mature-cell mean of the *drawn* reference arm. Let P be the eligible two-patient source pool, M_P its mature-cell count, and Y_j its unshifted target-gene counts. The population parameter of this finite empirical distribution is instead

$$\mu_P = \frac{1}{M_P} \sum_{j \in P, \text{ mature}} Y_j, \quad \theta_P(s) = f_N(s - 1)\mu_P. \quad (3)$$

We compute μ_P from all eligible source cells before either synthetic arm is sampled. Given P , the generator has $E[\tilde{Y}_N] = \mu_P$ and $E[\tilde{Y}_T] = s\mu_P$, making Equation 3 its true intrinsic mean contrast. It is fixed across cell draws from that pool. Patient-pair selection changes P ; reported coverage averages this conditional-pool question over the selected pairs. An all-patient pooled mean would use a different population from the generator’s two-patient draw. Neither target supplies inference over new patients.

The controlled eight-seed grid retains inclusion of Equation 2. A paired follow-up scores the same intervals against both equations at counts 5, 50, 100 and 800, with 200 replicates per cohort, pool and shift ($s = 0.5, 1$) and the first controlled-run seed. All 16 alternative settings reproduce the saved benchmark-inclusion, discrimination and width summaries; all 6,400 outputs are finite. This is a target sensitivity analysis, not an independent assessment of a newly selected rule. The maximum binomial Monte Carlo standard error is 3.54 percentage points; intervals below quantify Monte Carlo precision conditional on the observed source data.

2.2 Count grids change candidate cutpoints

Historical grids returned reference-only binned candidates of 100, 90 and 70, but changed random streams, bin boundaries and replicate budgets together (Appendix B). To isolate these choices, we simulated the union of counts once per cohort, pool and each of eight seeds, with 200 outer replicates and 200 inner bootstrap draws. Every grid summary reuses the same outputs at shared counts. The committed grid reaches $\{0, 20, 40, 100, 200, 400, 800\}$; filled and dense grids add count support. Fixed bins share boundaries across grids; adaptive bins recalculate them. Binned candidates are median counts within the selected bin, not estimated crossing locations.

With common fixed bins, SMC returns 100 on the committed grid and 70 on both expanded grids in all eight seeds. Adaptive bins return 65 on seven filled-grid seeds and 90 on one, while the dense grid returns 70 throughout. KUL3 instead returns 400 on six seeds and 800 on two, with all grid and binning schemes agreeing within each seed. Pooled draws yield no joint candidate in either cohort. The largest binomial Monte Carlo standard error in this grid is 0.035.

Table 1: Joint coverage-and-discrimination candidates from the controlled union-grid run. Entries give candidates with seed frequencies in parentheses where they vary; otherwise all eight seeds agree. Every grid summary reuses the same draws at shared counts. “None” means no tested count or bin qualifies.

cohort / pool	grid	fixed bins	adaptive bins	per count
SMC / reference	committed	100	100	100
	filled	70	65 (7), 90 (1)	50–80
	dense	70	70	45–80
KUL3 / reference	all three	400 (6), 800 (2)	same	same
Both / pooled	all three	None	None	None

At the committed count of 50, SMC reference-only discrimination ranges from 75.5% to 81.0% across seeds, with both criteria met on two of eight seeds. KUL3 reference-only discrimination is only 45.0–55.0% at that count. The reporting boundary therefore misses the discrimination requirement in six of eight SMC seeds and every KUL3 seed.

The per-count summaries in Table 1 supply a useful check on the bin medians. In SMC, the filled grid’s first qualifying actual count is 50 on two seeds, 60 on five and 80 on one; the dense grid adds a candidate of 45 on one seed. Thus a stable binned candidate of 70 is not evidence that 70 is a uniquely determined biological threshold. A bin pools outcomes over several count settings, whereas a per-count rate addresses only that setting. Fixed bins hold the boundaries constant, but adding settings can still change both a bin’s median count and its aggregate performance.

2.3 Fixed-pool coverage is lower at the reporting boundary

At 50 cells, fixed-pool coverage is 86.5% and 91.0% for SMC pooled and reference-only draws, and 87.0% and 84.5% for KUL3. The corresponding 95% Wilson intervals are [81.1,90.6], [86.2,94.2], [81.6,91.0] and [78.8,88.9]%, respectively. The first three leave uncertainty about the 90% requirement; KUL3/reference is below it even at the upper endpoint. With discrimination of 52.5%, 78.0%,

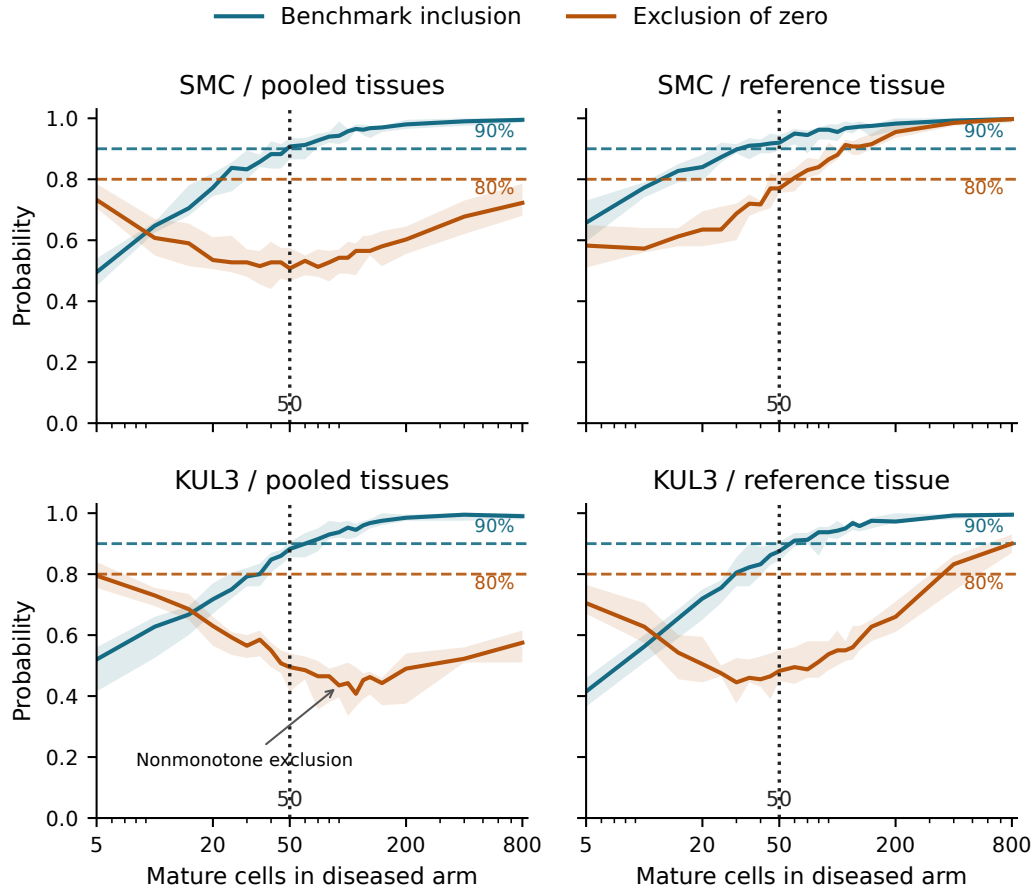


Figure 1: Direct evaluation on the controlled union of counts, without binning. Rows show cohorts; columns show draw pools. Curves are median rates over eight seeds and bands span their minima and maxima, conditional on the observed cohorts. Dashed lines mark the 90% benchmark-inclusion and 80% discrimination targets; the vertical line marks the committed count of 50. Rates condition on defined intervals and coverage uses the draw-dependent reference in Equation 2. Zero-cell settings have no defined interval and are omitted. No pooled-cell candidate qualifies. Per-count reference-only candidates vary by seed and cohort (Table 1).

123 54.0% and 55.0%, none of the four 50-cell conditions meets both empirical criteria in this seed
124 stream.

Table 2: Paired evaluation of the same nominal 95% intervals at $s = 0.5$. Entries are drawn-reference inclusion (D) and fixed-pool coverage (F), in %. Each entry uses 200 finite replicates; only the evaluation target changes.

cohort / pool	5 cells		50 cells		100 cells		800 cells	
	D	F	D	F	D	F	D	F
SMC / pooled	48.5	52.0	91.5	86.5	96.0	89.5	100.0	94.0
SMC / reference	67.5	67.5	92.5	91.0	92.5	90.0	100.0	90.5
KUL3 / pooled	54.5	60.5	87.0	87.0	97.0	91.0	99.0	91.0
KUL3 / reference	40.5	45.5	86.5	84.5	93.5	89.5	99.5	92.5

125 Table 2 separates target choice from changes in the estimator or interval. At 800 cells, drawn-reference
126 inclusion is 99.0–100.0%, but fixed-pool coverage is 90.5–94.0%. The paired decreases are 6.0, 9.5,
127 8.0 and 7.0 percentage points for SMC/pooled, SMC/reference, KUL3/pooled and KUL3/reference;
128 their Monte Carlo standard errors are 1.68, 2.08, 1.92 and 1.81 points. At five cells, switching targets
129 can instead raise inclusion. Thus the target change has a measurable effect whose direction depends
130 on sample size; neither score can substitute for the other.

131 The increasing discrepancy at large tumour-arm counts has an analytic explanation. The estimate
 132 is $\hat{\theta} = f_N(\bar{Y}_T - \bar{Y}_N)$, whereas its error against the drawn benchmark is $\hat{\theta} - i_{\text{req}} = f_N(\bar{Y}_T - s\bar{Y}_N)$.
 133 For independent arm draws conditional on P , their error variances differ by

$$\text{Var}(\hat{\theta} - \theta_P | P) - \text{Var}(\hat{\theta} - i_{\text{req}} | P) = f_N^2(1 - s^2) \frac{\sigma_P^2}{n_N}, \quad (4)$$

134 where σ_P^2 is the source-pool expression variance and $n_N = 800$. At $s = 0.5$, the drawn benchmark
 135 removes three quarters of the reference-arm variance contribution from the error being assessed,
 136 while the bootstrap continues resampling both arms. This explains why near-perfect benchmark
 137 inclusion can accompany lower coverage of the generating parameter. Skewed finite-sample errors
 138 still determine the actual percentile-interval rates.

139 The fixed-pool analysis covers four counts and one seed stream. We therefore retain the eight-seed
 140 grid selections as benchmark-based results and do not reinterpret them as fixed-pool calibrated
 141 cutpoints.

142 2.4 Sparse cells inflate null rejection

143 At $s = 1$, both targets are exactly zero, so exclusion of zero is a false positive. Table 3 pairs null and
 144 effect rates for the same interval method. The five-cell entries inspect pre-gate arithmetic; the 50-cell
 145 entries evaluate the reporting boundary.

Table 3: Exclusion of zero (%) with the same percentile interval under a halving (effect, $s = 0.5$) and the null ($s = 1$). Each entry uses 200 finite draws from one seed stream; nominal null rejection is 5%. Diagnostic runs are exploratory. Five-cell intervals are evaluated before the reporting gate, which abstains below 20 cells.

cohort / pool	5 cells		50 cells		100 cells		800 cells	
	effect	null	effect	null	effect	null	effect	null
SMC / pooled	70.5	63.0	52.5	28.5	56.0	18.0	69.5	9.5
SMC / reference	58.0	37.5	78.0	12.0	88.5	7.0	98.5	7.0
KUL3 / pooled	75.5	67.0	54.0	30.0	33.5	19.0	53.5	4.5
KUL3 / reference	70.0	61.5	55.0	15.5	56.0	16.0	87.0	6.5

146 At five cells, null rejection is 37.5–67.0%, far above the nominal 5%. For the pooled SMC and KUL3
 147 alternatives, respectively 96 and 85 of 200 tumour samples have all-zero target expression; all of
 148 these exclude zero. Resampling such a tumour sample cannot recover its unobserved positive tail.
 149 The reference arm still varies, so none of the 6,400 full intervals has exactly zero width. The pooled
 150 five-cell median widths are 1.45 and 1.72, compared with 2.35 and 2.01 at 50 cells. The diagnostic
 151 supports sparse-sample undercoverage with variation still contributed by the reference arm. Even at
 152 50 cells, null rejection is 12.0–30.0%. The original 90% benchmark-inclusion criterion at $s = 0.5$
 153 therefore cannot substitute for checking the null.

154 2.5 First qualification can reverse at larger counts

155 Three of eight SMC/reference seeds in the complete saved union grid fail a criterion after first
 156 qualifying (Table 4). This matters for a rule that reports all counts above a scalar boundary: passing
 157 at its first point does not ensure performance over its reporting range. Requiring every subsequently
 158 evaluated point to qualify changes candidates 45, 50 and 50 to 60, 70 and 70. No reversal occurs in
 159 the eight KUL3/reference seeds; no pooled candidate qualifies. These are descriptive selections on
 160 the recorded benchmark.

Table 4: The three SMC/reference seed streams with a later failing count. The last column requires all subsequent evaluated counts through 800 to pass; it is a descriptive alternative to the first-crossing rule.

seed stream	first qualifying	later failing	all subsequent pass
3	45	50	60
6	50	60	70
7	50	60	70

2.6 Patient influence and the cohort gap

Enumerating all 45 and 15 two-patient holdout pairs separates patient omission from repeated simulation. Keying streams to the sorted pair identifiers keeps every retained pair fixed. Raising the inner bootstrap budget from 200 to 1,000 and 5,000 leaves the first cohort's candidate at 70, but moves the second's 400 to 300 and back to 400. Omitting one source patient changes the joint coverage-and-discrimination candidate for none of ten patients in the first cohort and three of six in the second (KUL01, KUL30 and KUL31). For the coverage-only candidate, the corresponding counts are five of ten and zero of six. These checks quantify influence within each observed cohort. Appendix B gives the biological comparability limits.

Table 5: Reference-only sensitivity with every two-patient holdout pair represented once per count and effect. The three inner budgets reuse identical outer samples. Omission counts compare with the baseline at 200 inner draws; all pairs involving the omitted patient are removed and retained pairs are unchanged. These are sensitivity summaries within the observed cohorts.

cohort	pairs	joint candidate, inner draws			omissions changing candidate	
		200	1,000	5,000	joint	coverage only
SMC	45	70	70	70	0/10	5/10
KUL3	15	400	300	400	3/6	0/6

The sensitivity windows cover counts 40, 50, 60, 70, 80, 90, 100, 110 in SMC and 200, 300, 400, 600, 800 in KUL3. These per-count summaries use a fixed pair schedule, so they are a separate experiment from the eight-seed union grid. Omitting KUL01 or KUL30 moves the joint candidate from 400 to 300; omitting KUL31 moves it to the lowest tested count, 200, with its lower boundary unobserved. The coverage-only baselines likewise occur at the lowest tested counts, 40 and 200. Their stability cannot locate a lower crossing outside those windows. Overlapping pairs do not add independent patients.

Expression dispersion partly explains the cohort gap. Holding tissue eligibility fixed at reference-only, the complete SMC and KUL3 mature-cell pools contain 662 and 844 cells. Across their 45 and 15 eligible patient pairs, median source-pool squared coefficients of variation ($c_P^2 = \sigma_P^2 / \mu_P^2$) are 4.35 and 16.99; median zero-expression fractions are 39.3% and 62.1%. The source means are 19.5 and 11.9 raw counts, respectively (medians over pools). The KUL3 contrast therefore has substantially more relative sampling noise despite its larger source-cell pool.

For the implemented binomial thinning, the tumour-cell variance is $s^2\sigma_P^2 + s(1-s)\mu_P$. At $s = 0.5$, the exact conditional variance of the contrast relative to its squared signal is

$$\frac{\text{Var}(\hat{\theta} | P)}{\theta_P(0.5)^2} = \frac{c_P^2 + 1/\mu_P}{n_T} + \frac{4c_P^2}{n_N}. \quad (5)$$

With $n_T = 50$ and $n_N = 800$, the median signal-to-standard-deviation ratios are 3.02 in SMC and 1.53 in KUL3. This provides a distributional explanation for weaker KUL3 discrimination. It does not reproduce the exact 70-versus-400 candidate ratio: percentile-bootstrap behavior, pair heterogeneity and binning also affect that selection. The biological causes of the source-distribution difference remain unexplained.

2.7 Biological controls fail to establish mechanism

The pre-committed panel included compositional, intrinsic and retained-gene controls. In the primary cohort the retained-gene control changed by -1.000 to -0.951 relative to baseline, almost matching the compositional controls' -1.000 to -0.952 . The intrinsic control had only 0.04 CP10K, compared with 19.4 and 24.9 for the abundant compositional controls (counts per 10,000), making it uninformative for the intended test. A second cohort reproduced the pattern. Thus the application has two distinct unresolved problems: its interval rule lacks the recorded calibration support, and its control panel does not distinguish the proposed biological mechanisms. Improving one would not establish the other.

3 What a replacement validation would need to establish

The paired analysis shows why specifying the target changes an assessment: the same intervals can have different inclusion rates against a sampled reference and the generating-pool parameter. A clinical simulator likewise needs to distinguish a requested population treatment effect, a latent finite-cohort effect and an observed-data contrast. Coverage must use the target for which the interval was designed.

The earlier recovery curve admits a simpler explanation. The oracle estimate applies Equation 1 to the sampled fractions and means, exactly reproducing their empirical decomposition. This arithmetic identity explains the comparison without evaluating its uncertainty. For example, the same sample mean has about 95% or 36% coverage when only its interval width changes (Appendix C). Established trial-estimator relationships and learned-generator checks are collected there as supporting examples.

Assess the whole decision rule using the interval that would actually be reported. Record bias and error against the chosen target, null rejection, detection at the specified effect, and abstention frequency. Report both conditional interval performance among returned answers and how often a valid answer is returned. Our five-cell diagnostic concerns arithmetic before a gate that already abstains there; the 50-cell null results concern its reporting boundary directly. A successful effect-detection rate can accompany excessive null rejection, so the former alone cannot validate an interval. A fixed 5% null target is appropriate here because the reported interval is nominally 95%; other operating targets must follow the intended use.

Separate selecting a rule from assessing it. If a scalar threshold is intended to guarantee performance above that threshold, qualification must address the evaluated upper range rather than only its first passing point. Reserve independent simulation draws for the final assessment, report rate uncertainty and the tested range, and distinguish this from transfer to new patients or cohorts. Increasing resamples from an empirical pool improves conditional numerical precision; it does not broaden that pool's biological support. For an external-control application, the additional question is comparability across data sources, which requires direct study-level validation such as the held-out comparisons of Ventz et al. [2019]. Evaluating transfer to patient populations requires that additional evidence.

3.1 A synthetic external-control example: validating the wrong population

A separate constructed experiment makes target-population mismatch consequential. A fixed synthetic registry supplies a single-arm trial and three external controls per enrolled patient. Enrolment favors heavily pretreated strata; induced benefits vary by stratum and latent individual responsiveness. The intended target is the mean individual treatment effect among enrolled records. We compare stratum contrasts standardized to the enrolled population with the same contrasts averaged over the pooled trial-plus-control file. The latter estimates a different population effect; it is defective only relative to the stated enrolled-population target. Appendix D specifies the design and controls.

Table 6: Reference choice reverses the estimator ranking in the synthetic external-control example. Bias and RMSE use 200 studies with 1,600 enrolled records and 4,800 controls. Columns identify the population used to average observed stratum contrasts; only enrolled standardization matches the intended target. All entries are in abstract outcome units.

Validation reference	Pooled standardization		Enrolled standardization	
	Bias	RMSE	Bias	RMSE
Observed contrast, pooled	0.000	0.000	-0.872	0.876
Potential outcomes, pooled	+0.003	0.115	-0.869	0.875
Potential outcomes, enrolled	+0.885	0.892	+0.013	0.105

At 1,600 enrolled patients, pooled standardization has a median estimate of 1.890 outcome units while the enrolled records' median true effect is 1.003. These medians fall on opposite sides of the prespecified illustrative threshold of 1.5; this is not an estimated probability of a wrong decision. Against the observed pooled contrast, the pooled estimator has zero error by construction, while enrolled standardization appears inaccurate (Table 6). Using the enrolled potential-outcome reference reverses that ranking. Merely using potential outcomes is insufficient: averaging them over the pooled

population still favors the estimator of the wrong target. The target must specify both the effect and the population over which it is averaged.

Two controls limit the interpretation. At zero case-mix shift, pooled standardization still reproduces its observed-data reference but its bias against the enrolled effect is only -0.002 units. Equality therefore does not establish that an estimate is wrong. Under the full shift, pooled weighting balances the measured strata between arms to numerical precision while their weighted distribution differs from the enrolled distribution (maximum standardized difference 0.964). Between-arm balance does not establish target-population alignment. This constructed, uncensored outcome experiment demonstrates a possible validation failure; it supplies no evidence of its prevalence or of performance in an actual clinical trial.

4 Conclusion

The rule's reporting boundary has inadequate null behavior, and its fixed-pool coverage estimates are 84.5–91.0% at that boundary in the paired diagnostic. Near-perfect inclusion of the sampled reference at larger counts overstates coverage of the generating-pool effect. The variance calculation identifies why changing the target changes this assessment. Separately, first-crossing candidates can fail at larger counts, and cohort differences in relative expression variance help explain the different detection curves.

The evidence establishes a failure in this single-cell application. A replacement rule needs a specified target, joint assessment of null rejection, effect detection and abstention, and independent evaluation after selection. The synthetic external-control example shows why an independently constructed reference must also average over the intended target population. The broader patient-population question requires new patients and cohorts.

5 Responsible use, limitations and impact

Misuse and limitations. NOT ESTIMABLE must not be read as “no effect” or as evidence that surviving cells are behaving normally. A separate within-cell analysis over three cohorts and 42 patients also remained unresolved because its comparability checks returned *undecided* (Appendix B). Calibration here covers one gene, one labelling axis and one weighting on two small cohorts. It challenges this rule's justification, not cell-count rules in general. The fixed-pool analysis estimates coverage conditional on empirical cell distributions; uncertainty over new patients remains unmeasured. The draw-pool choice was underspecified, repeated simulations do not add independent patients, and the leave-one-patient-out sensitivity remains conditional on these cohorts; patient-level population uncertainty in the candidate cutpoints is unquantified. The absence-defined labelling axis remains sensitive to sequencing depth (Appendix B). The external search found no affirmative case, so the documented failure remains a self-audit rather than evidence about its prevalence. The external-control example uses a synthetic registry and an illustrative decision threshold; it does not establish clinical benefit or support a treatment decision.

Data and impact. The single-cell cohorts are public, de-identified and previously published. The release policy covers derived summaries and code, rather than cell-level matrices. A system reporting group-specific quantities should distinguish an unavailable comparison from an estimated null and should evaluate the coverage and discrimination of any abstention rule. Publishing failed calibration attempts helps prevent unsupported thresholds from being treated as validated.

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A Simulation settings

Experiment	Fixed settings and sampling units
Single-cell calibration	GSE132465 (SMC): 10 paired patients; GSE144735 (KUL3): 6. Target: GUCA2A; <code>stem_pole</code> axis, <code>lineage</code> resolution, <code>differentiated</code> mature label; reference (<code>normal</code>) weighting. Each replicate holds out two patients and draws 2,000 cells per synthetic arm from their eligible pool; reference mature fraction 0.40. Expression shifts are 1.0, 0.8, 0.5, 0.25; calibration uses 0.5.
Interval construction	A nominal 95% percentile interval from 200 resamples of mature cells within each synthetic arm, holding cell fractions fixed. This assesses the conditional intrinsic contrast; it does not estimate population uncertainty across patients. Coverage and discrimination condition on defined intervals. The historical inclusion score uses the drawn reference in Equation 2; the paired follow-up also evaluates fixed-pool coverage using Equation 3.
Patient outcomes	Stratum probabilities (0.5, 0.3, 0.2), baseline means (10, 14, 25) and $Y \mid (g, A) \sim \mathcal{N}(\mu_g + 3A, 4^2)$. Confounded treatment probabilities (0.25, 0.50, 0.75); alternatives are independent Bernoulli assignment at 0.5 and complete balanced blocks of size 4 within strata. The three-design sweep uses 200 replicates per size and seed across six seeds.
Survival outcomes	The same stratum and confounded-assignment probabilities; exponential event rate $\lambda_g \exp(\beta A)$, $\lambda = (0.10, 0.16, 0.30)$, $\exp(\beta) = 0.6$. Administrative horizons 1000, 20.065216, 6.072920, 2.028858 target 0%, 12%, 49%, 78% censoring. The same latent cohort is viewed through each horizon; RMST uses $\tau = 2$. Each size has 200 replicates per seed over six seeds.

Historical grid budgets are reported in Appendix B; Section 2 specifies the controlled runs and paired target analysis. The source cells are observed data; the treatment and survival outcomes are simulated.

B Calibration details and limits

Historical grids. Before the controlled run, the SMC reference-only sweeps returned binned joint candidates of 100, 90 and 70 on the committed, filled and dense grids, with 500, 500 and 200 replicates per setting, respectively. Their adjacent crossing brackets were (40, 100], (65, 90] and (42.5, 70], describing evaluated bin summaries rather than confidence intervals. The 100-to-90 change occurred within the same bin, (76, 107]: adding counts 80 and 90 to 100 changed both its median count and its aggregated outcomes. Random streams and bin boundaries also differed, so this comparison alone did not isolate a resolution effect. The controlled analysis in Section 2 supersedes it for that question. Under pooled draws no joint cutpoint qualified in either cohort’s historical grids. KUL3 reference-only candidates were 400 or 800; its largest setting sampled 800 mature cells per arm with replacement from 844 distinct mature source cells. Sampling with replacement is a valid empirical-pool simulation, but an 800-draw arm is not 800 distinct cells or additional independent patients.

Recovery accounting and unavailable contrasts. The four diagnostic sweeps comprise 1,711,200 replicates: 1,283,400 with nonzero requested intrinsic reference and 427,800 with zero requested reference, for which ratios are undefined. The first 50-replicate sweep has 119,600 draws, partitioned into 89,700 and 29,900, respectively. The maximum estimate–realised-reference residual is zero throughout. At a zero-cell setting the raw ratio is $1/(1 - s)$ when the denominator is nonzero. These raw diagnostic values arise from the harness setting an empty-arm mean to zero; they do not define an effect in absent cells. The reporting gate abstains and the interval is undefined. Separately, undefined intervals were previously counted as failures through comparisons with NaN; corrected rates exclude them and report abstentions. An all-abstaining bin has no coverage estimate. One historical seed gives $ok = wide = 45$, showing that a three-state specification need not yield three distinct fitted regions.

Biological scope and patient uncertainty. The finest annotation resolution yields abstention in 28/28 patients under one population definition and 24/28 under the other; coarser definitions do not resolve identifiability. The coarsest definition labels every epithelial cell mature and forces the

compositional term to zero. In a separate comparability analysis over three cohorts and 42 patients, 83 of 84 housekeeping-detection control rows pass a 0.10 tolerance despite expression changes of 1.19–1.64 fold. Patient-bootstrap $\log_2$ expression intervals cross the accepted $[-0.5, 0.5]$ boundaries in every cohort, leaving the comparisons undecided. Depth matching discards roughly three quarters of the cells and pushes about half the remaining estimable rows below the count threshold. Within SMC, changing tissue eligibility changes the mature-cell pool from 662 to 3,824 cells. The across-cohort comparison in Section 2.6 instead holds tissue eligibility fixed. These observations restrict biological interpretation. Repeated cell draws and overlapping patient pairs do not supply independent patient replication.

C Supporting estimator and trial checks

Reference dependence. With nonempty stratum arms, saturated G-computation [Robins, 1986], IPW using empirical stratum propensities, non-cross-fitted saturated AIPW and matching using every control all reproduce $T(D) = \sum_g (n_g/n)(\bar{Y}_{1g} - \bar{Y}_{0g})$. The IPW weights recover each stratum mean; the AIPW augmentation sums to zero within stratum. OLS with additive stratum indicators instead weights contrasts by normalized $n_g \hat{p}_g(1 - \hat{p}_g)$ [Angrist, 1998, Frisch and Waugh, 1933]. Under confounded assignment at 5,000 patients, switching between these two references changes the G-computation/OLS residuals from 0/0.0663 to 0.0663/ 3.5×10^{-14} . Complete 1:1 blocks make the weights coincide. These are implementation checks of established identities. Under effect heterogeneity, OLS recovery rises from 1.003 to 1.132 while its own-reference residual stays at numerical precision.

Interval control. For 200 independent $\mathcal{N}(\theta, 1)$ observations and 2,000 draws per effect, the full-sample mean has 94.60% and 95.40% coverage at $\theta = 0, 0.5$. Quartering the interval half-width leaves the estimate and its zero residual against the empirical mean unchanged, but reduces coverage to 35.95% and 36.30%. The control illustrates why residuals cannot certify intervals.

Censoring and implementation checks. The standardized exponential estimator equals its events/exposure reference. At 3,000 patients its maximum departure from the latent-time reference grows from zero without censoring to 0.376 at 78% censoring. RMST at $\tau = 2$, below every censoring horizon, agrees with the empirical latent truncated-mean contrast to 2.8×10^{-15} . The most affected setting excludes 463 of 1,200 pairs with nonfinite fits. Eight independent Cox comparisons agree with lifelines to 1.4×10^{-7} against a 10^{-5} tolerance.

Learned-generator design and exclusions. Seven generators (four Gaussian mixtures, an MLP conditional Gaussian and two parametric controls), ten estimators, six sizes, six seed streams and 30 replicates schedule 75,600 evaluations on 7,560 cohorts. Generator failures leave 75,580 evaluations on 7,558 cohorts; 74,726 drawn-reference pairs are finite. The 854 exclusions comprise 410 nonfinite references and 731 nonfinite estimates, overlapping in 287 evaluations. Saturated G-computation has zero residual in every setting because the identity holds on each drawn dataset. Ridge and MLP outcome fits can depart; the largest mixture-generated MLP residual is 3.3. Generator complexity by itself does not test an estimator identity or establish fidelity to patients.

Tolerance and reference choice. Of 420 settings, 41 ridge settings pass only a relative-plus-absolute comparison. Thirteen generator–estimator combinations switch into this category between $n = 100$ and 5,000. The comparisons use 10^{-8} against zero and $10^{-8} + 10^{-5}|T|$ against the reference. Report raw maxima, outcome units and both tolerances; distinguish a proved identity from agreement on tested draws. The optional ratio of median departure to median reference error is undefined at zero denominator and cannot rank estimators. Mixture model-target references use 200,000 Monte Carlo observations; saved median standard errors are 0.019–0.027. Parametric and MLP model-target references are analytic.

D Synthetic external-control design

The fixed registry contains 50,000 simulated records with stratum shares (0.55, 0.30, 0.15), baseline outcomes $Y_0 \mid g \sim \mathcal{N}(\mu_g, 3^2)$, $\mu = (12, 8, 4)$, and effects $Y_1 - Y_0 = \delta_g r$, $\delta = (3, 1.5, 0.25)$,

447 $r \sim \text{Gamma}(4, 0.25)$ (shape, scale). Responsiveness is fixed per record and hidden from estimators.
448 Enrolment weights favor shares (0.15, 0.25, 0.60); both arms sample without replacement and are
449 disjoint. The pooled target is a mixture of enrolled and control records, not the registry population
450 itself. Normal outcomes can be negative: this is an abstract continuous-outcome experiment, not a
451 fitted progression-free-survival model.

452 There are 200 study draws at each of $n = 50, 100, 200, 400, 800, 1600$, using one seed (20260915)
453 and one fixed registry: 1,200 studies, eight estimators and five references yield 48,000 comparisons:
454 47,969 finite values and 31 exclusions from one study with an empty stratum arm at $n = 50$. The
455 registry uses the same seed. The intended finite-cohort truth averages both potential outcomes over
456 enrolled records; it is available to the simulator, not to estimators observing one outcome per record.
457 Five case-mix shifts and 50 balance draws at $n = 1600$ provide controls. Threshold crossings in the
458 saved shift table compare medians, not replicate-level decisions.

459 For enrolled standardization, RMSE against enrolled truth falls from 0.608 at $n = 50$ to 0.105 at
460 $n = 1600$; pooled standardization falls from 1.082 to 0.892. The nine prespecified falsification
461 checks did not fire, but this shows consistency with those checks, not clinical realism. The example
462 assumes the same target-population mistake in estimator and reference; its identical recovery curve
463 may itself alert a careful analyst. It neither estimates a failure frequency nor requires a reference to
464 disagree numerically with a correct estimator on every dataset.

465 E External audit and research context

466 **Search and classification.** A pre-registered queue contained 41 entries; 19 were screened out and
467 seven classified within the 26 entries examined in order. An amendment before code inspection
468 relaxed unsuccessful queries; the classification rule was unchanged. Generating-parameter references
469 or quantities computed outside the estimator’s observed records give NO. Same-record references
470 require examination of inputs, weights, conditioning, nuisance fitting and observation process; YES
471 requires affirmative evidence of functional identity. All seven recorded verdicts are NO: *caus1*, *target-*
472 *trial-emulation*, *ecasim*, *PlasmodeSimulation*, *StatsPAI*, *diff-diff*, and *Auto-Empirical-Research-*
473 *Skills*. The first six use generating parameters or outside-sample quantities; the last compares
474 cross-fitted partialling-out with full-sample OLS. Its 4.0×10^{-14} residual on noiseless linear data
475 becomes 0.125 after adding outcome noise. The functions agree on the restricted noiseless support but
476 not on general noisy datasets; the NO verdict concerns shared functional form. Classification used one
477 reader, only this last repository was executed, and the sample is nonrandom. The virtual-control-arm
478 query located no examinable code. These limits preclude a prevalence estimate.

479 **Related scope.** Cell-count filters are established in pseudobulk analysis [Crowell et al., 2020];
480 scCODA, Milo, propeller and DA-seq address composition [Büttner et al., 2021, Dann et al., 2022,
481 Phipson et al., 2022, Zhao et al., 2021], while cacao separates composition and expression changes
482 [Petukhov et al., 2022]. Selective prediction motivates evaluating both answers and abstentions
483 [Chow, 1970, El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]. Conformal prediction
484 coverage [Vovk et al., 2005, Angelopoulos and Bates, 2023] does not define a missing conditional
485 population or automatically provide intervals for this intrinsic parameter.

486 **Documentary source of the case.** The anonymous review record includes the original validation
487 statement (Artifact A), interval criteria (Artifact B), and numerical context. Exact source mapping
488 and checksums remain outside the review bundle. This records one internal validation error; the
489 external search found no positive case.

490 **Implementation and provenance.** Corrections separated collapsed population definitions, aligned
491 labelling, tracked producing scripts, and repaired undefined-interval denominators. Saved tables
492 retain commit hashes and seeds; these establish traceability, not statistical validity.

493 **Use of large language models.** Language-model assistants supported coding, figures, drafting and
494 model audits under author direction. They helped identify defects and introduced some of those
495 above. Authors remain responsible for all claims. Automated checks re-derive selected results and
496 complement review; they do not verify every manuscript statement.